

Do Temporal Network Signals Predict How Scientific Concepts Spread Across Disciplines?

Internal Research Report

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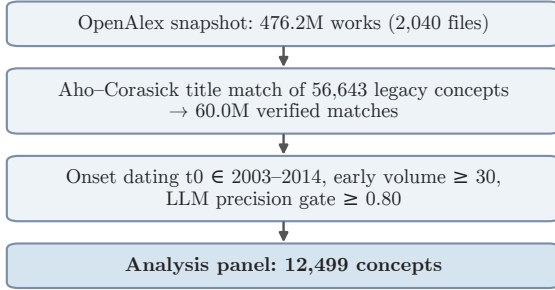
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1 Overview

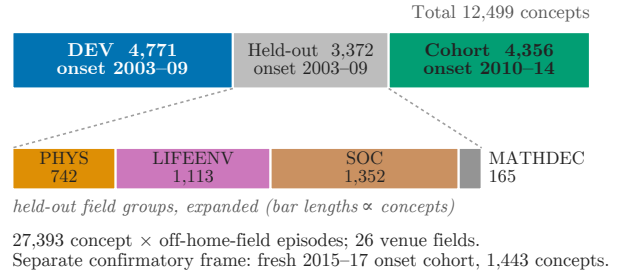
This report is the chronological record of a five-iteration research run investigating whether the early structure of a scientific concept’s topic co-occurrence network anticipates its later cross-disciplinary breadth. The run used the OpenAlex bulk snapshot (476,196,327 works; 2026-09-23) to identify 12,499 concepts, track 27,393 adoption episodes across 26 fields, and screen 53 early network indicators against a five-feature popularity baseline. The study also tested whether concepts spread next to fields related to the ones currently retaining them, extending the principle of relatedness from economic complexity.

Each section below corresponds to one iteration, in the order it ran. Earlier sections are not rewritten; they are the record of what was believed at the time. Dead ends are kept and labelled. The closing section draws together what the run learned and what remains open.

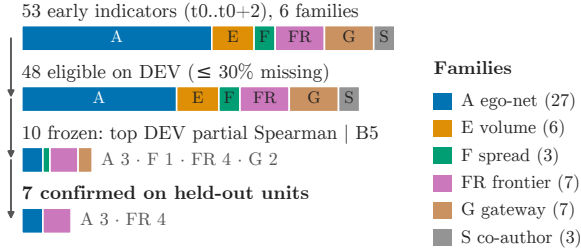
(a) Concept identification



(b) Panel structure



(c) Indicator screen (breadth outcome)



(d) Research questions

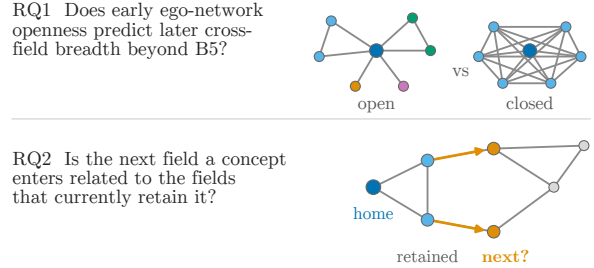


Figure 1: Overview of the study design. (a) Concept identification: 56,643 legacy OpenAlex concepts are matched by title (Aho–Corasick) against a 476.2M-work OpenAlex snapshot, giving 60.0M verified matches; concepts are onset-dated ($t_0 \in 2003-2014$), required to have early volume ≥ 30 and to pass an LLM precision gate ≥ 0.80 , yielding an analysis panel of 12,499 concepts. (b) Panel structure, bar lengths proportional to concept counts: DEV (blue; CS, Eng, BGM and Med homes, onset 2003–09) 4,771, held-out field groups (grey; onset 2003–09) 3,372, and the 2010–14 onset cohort (green) 4,356. The expanded lower bar splits the held-out block into PHYS 742, LIFEENV 1,113, SOC 1,352 and MATHDEC 165. The panel contains 27,393 concept $\times$ off-home-field episodes over 26 venue fields; a separate fresh 2015–17 onset cohort of 1,443 concepts serves as a confirmatory frame. (c) Indicator screen for the breadth outcome (O2r_m50). Bars are coloured by indicator family with lengths proportional to indicator counts: 53 early indicators computed over $t_0..t_0+2$; 48 eligible on DEV; the 10 frozen on DEV by partial Spearman correlation given the five-feature baseline B5; and the 7 confirmed on held-out units. (d) The two research questions, drawn as schematic illustrations with no data.

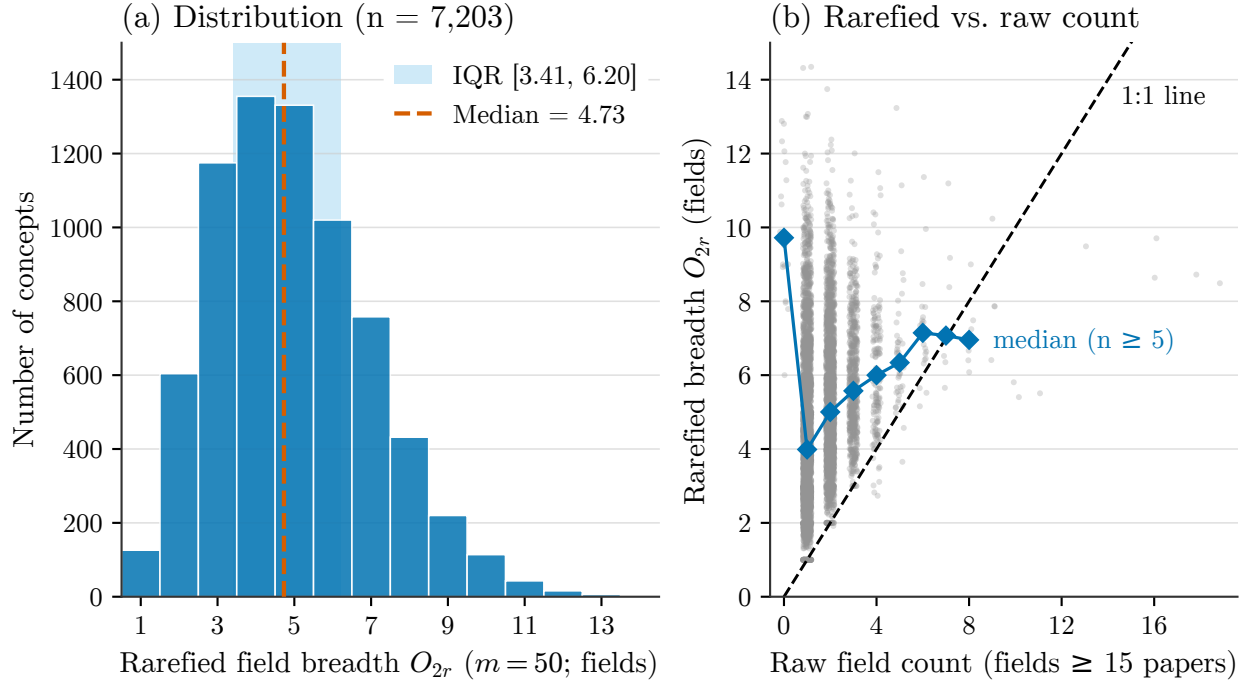


Figure 2: Distribution of the primary outcome, rarefied field breadth (O_{2r} , $m = 50$), over the 7,203 of 12,499 frame concepts that have at least 50 papers in the $t_0+6..t_0+8$ outcome window. (a) Histogram of O_{2r} in 1-field bins centred on the integers. The dashed orange line marks the median (4.73) and the light-blue band the interquartile range [3.41, 6.20]. (b) O_{2r} against the raw field count. Rarefied breadth lies above the 1:1 line for 97.6% of concepts and is nearly independent of outcome-window volume ($\rho = -0.06$), whereas the thresholded raw count grows with volume ($\rho = 0.41$).

2 Iteration 1: Wide Screen of Five Rival Mechanisms

2.1 Why this iteration ran

The run began with five rival mechanisms for why new concepts become broadly integrated: (1) lineage naturalisation (background-adjusted A_h^*), (2) social reach through unconnected author groups (candidate S), (3) structural diversity of new co-occurrence neighbours (candidate D), (4) frequency-free selectivity (candidate F), and (5) landing in gateway fields plus adopter insularity (candidate G). All five were screened on one frozen DEV panel, with one baseline, one outcome set and one pre-registered rule. The strategy was motivated by the hypothesis-stage reviewer’s concern about low reliability (~ 0.32) for A_h^* from the probe.

Five artifacts were commissioned; three completed. Two failed (gen_art_dataset_1, the sealed held-out test set, and gen_art_experiment_2, candidate S): both stalled during execution, so no held-out evaluation existed and candidate S was untested. The review scored the iteration at 3 (blocking).

2.2 Experiment 1: Naturalisation gap A_h^* (iteration 1)

This artifact screened the background-adjusted, field-stratified, partially pooled lineage naturalisation gap against the common B5 count baseline on 48 DEV concepts.

Table 1: Experiment 1 screen result: naturalisation gap A_h^* on the P78 DEV panel (gen_art_experiment_1, iteration 1).

Metric	Value
n used	48
$\Delta\rho$ (B5+ A_h^* vs B5)	−0.006
90% CI	[−0.034, +0.017]
ρ_{B5}	0.834
Positive groups (of 4)	0
Split-half reliability (SB)	0.583
$ \rho $ with log volume	0.145
$ \rho $ with growth	0.177
Survives pre-registered rule	No

The candidate failed all clauses except the size-correlation check. Clause results: $\Delta\rho \geq 0.10$ and CI low > 0 : FAIL (−0.006, [−0.034, +0.017]); positive groups $\geq 3/4$: FAIL (0/4); reliability ≥ 0.6 : FAIL (0.583); $|\rho|$ with volume/growth ≤ 0.6 : PASS (0.145 / 0.177).

An additional finding from this artifact: the R^2 of the raw lineage log-odds ratio on the background log-odds ratio was 0.66 (on the unthinned data, 0.72). The background log-odds ratio was positive for all 48 concepts: every field’s citing papers preferentially cite their own field above chance. Two-thirds of the between-concept variance in raw lineage autonomy is general disciplinary homophily.

Data deviation. The shared OpenAlex credit pool ran dry (139 credits spent). Yearly counts used OpenAlex S0 exactly; field labels, concept papers, citation lineage and background-reference fields came from free Semantic Scholar data (fractional s2-fos text-classifier fields). The S2 and OpenAlex O_{2r} agreed with Spearman 0.87 on 11 concepts.

2.3 Experiment 3: Co-occurrence diversity D and selectivity F (iteration 1)

This artifact screened two co-occurrence emergence indicators on the frozen P78 DEV panel (47 DEV concepts: BIO 16, CS 12, MED 10, ENG 9) under shared protocol S0.

Table 2: Experiment 3 screen result: co-occurrence diversity D_{ratio} and selectivity F_{res} (gen_art_experiment_3, iteration 1).

Candidate	$\Delta\rho$	90% CI	Groups+	SB	ρ_{vol}	Survives
D_{ratio}	+0.006	[−0.092, +0.135]	3/4	0.83	0.109	No
F_{res}	−0.060	[−0.158, +0.014]	1/4	0.44	0.037	No

Neither candidate survived the pre-registered rule. The D_z literal was dropped by the T3 size diagnostic (ρ with log volume = −0.63), so the pre-declared fallback D_{ratio} became the primary D. The backbone used full-corpus topic PMI per slice (2000–04 / 05–09 / 10–14), Leiden $\gamma = 3$ (25/26/23 communities).

Exploratory: the out-of-group partial ρ of D_{ratio} given B5 was 0.335 [0.02, 0.65], permutation $p = 0.037$; $\Delta\rho$ was near its ceiling because B5 was already strong ($\rho_{B5} = 0.770$). The portability table (34 indicators) showed D_{ratio} , D_{rare} , participation and NOV_{res} associated with O_{2r} in all 4 groups (pooled ρ 0.45–0.63), but redundant under $\Delta\rho$. Edge persistence showed a portable negative signal (within-group ρ : −0.34, −0.52, −0.07, −0.22).

2.4 Experiment 4: Gateway landing G (iteration 1)

This artifact was also the authoritative producer of the shared outcome tables (78 rows). It screened candidate G (the adopting field’s eigenvector gateway centrality on the 1998–2002 backbone) on 46 DEV concepts (34 with an outcome window).

Table 3: Experiment 4 screen result: gateway landing G on rarefied breadth O_{2r} (gen_art_experiment_4, iteration 1).

Metric	Value
$\Delta\rho$ (B5+G vs B5, O_{2r} m=30)	+0.033
90% CI	[−0.095, +0.168]
ρ_{B5}	0.327
Positive groups (of 4)	2
Reliability (SB)	0.92
Survives pre-registered rule	No
$O_{2r\text{resid}}$: $\Delta\rho$	+0.150
$O_{2r\text{resid}}$ 90% CI	[0.000, +0.321]
O_1 ΔAUC	+0.072
Field-level retention ΔAUC	+0.103
Field retention 95% CI	[0.034, 0.167]

G did not survive the pre-registered rule, although reliability was high. The field-level retention signal ($\Delta\text{AUC} +0.103$) was the strongest secondary result in the iteration and became the lead for iteration 2. However, the evaluation later showed this signal did not replicate beyond the 80-episode DEV panel (see Section 3.4).

2.5 Dead ends and failed artifacts (iteration 1)

- **gen_art_dataset_1** (sealed held-out test set): failed (REPL turn stalled). No held-out evaluation existed after iteration 1.
- **gen_art_experiment_2** (candidate S, unconnected co-author groups): failed (REPL turn stalled). Candidate S was untested until iteration 3.
- A_h^* : null ($\Delta\rho = -0.006$, 0/4 groups). Within-field sign flipped (Med +0.45 correlation with O_{2r} , CS -0.18). Closed as a headline bet.
- D_{ratio} : null under $\Delta\rho$ (+0.006). Exploratory partial ρ 0.335 [0.02, 0.65] was marginal with 1 of 12 partials having CI90 excluding zero.
- F_{res} : null (-0.060, reliability 0.44).
- **G (concept-level)**: null on primary ($\Delta\rho = +0.033$, 2/4 groups).

2.6 Hypothesis update after iteration 1

The headline moved from the concept-level naturalisation gap (A_h^* , null) to the field-level gateway-retention lead ($\Delta\text{AUC} +0.103$, DEV only, $n = 80$ episodes). The unit of analysis became the concept $\times$ off-home-field adoption episode, backed by REML variance decomposition ($\tau_{cj} = 0.65 > \tau_c = 0.29$). Evidence state: lead. Confidence: decreased. The review scored the iteration at 3 (blocking) and flagged that several conclusions contradicted the artifacts.

3 Iteration 2: Scaling Up and Testing Gateway Retention

3.1 Why this iteration ran

Iteration 2 set out to confirm or kill the gateway-retention lead from iteration 1, by building a much larger panel from the free OpenAlex S3 snapshot (zero API credits). Five artifacts were commissioned: Experiment 5 (decisive held-out test of gateway, building the authoritative 12,499-concept frame), Experiment 6 (field-entry conditional logit and trajectories), Evaluation 1 (immediate stress test of the iteration-1 lead), Dataset 2 (external-recognition lookup), and Research 1 (related-work positioning).

3.2 Experiment 5: Held-out test of gateway centrality (iteration 2)

One zero-credit scan of all 2,040 OpenAlex S3 works files (476,196,327 works; 129.4M base works 1995–2022). Concepts identified by Aho–Corasick title matching of 56,643 legacy concepts (levels 2–5) plus Wikidata aliases, with stemmed verification: 60.0M verified matches. The grounding rule TAG gave test precision 0.947, recall 0.659.

H1 result: DISCONFIRMED. The held-out ΔAUC of gateway centrality over the full covariate set was -0.00001 [-0.0006 , $+0.0003$]. DL pooled -0.00004 ($I^2 = 0$). The baseline ladder showed that gateway’s DEV signal ($+0.0019$) vanished once the field’s leave-concept-out retention propensity $P_{j(-c)}$ was added, and reversed on held-out (-0.0016). Gateway alone had AUC 0.605 on DEV vs 0.506 on held-out (0.41 in SOC): gateway was a domain-specific proxy for “fields that keep things”.

Table 4: Panel composition from Experiment 5 (gen_art_experiment_5, iteration 2).

Split	Concepts	Episodes
DEV (CS/Eng/BGM/Med, onset 2003–2009)	4,771	9,079
Cohort (onset 2010–2014, all fields)	4,356	9,799
Held-out Physical Sciences	742	1,662
Held-out Life & Environment	1,113	3,099
Held-out Social Sciences	1,352	3,320
Held-out Math & Decision Sci.	165	434
Total	12,499	27,393

H3 result (concept-level gateway $\rightarrow$ breadth): Held-out partial ρ : G 0.030, G_A 0.026, G_{btw} 0.046 (Holm $p = 0.0045$); DL pooled G 0.068 [0.029, 0.107]. The effect was small, with DEV-to-held-out shrinkage from 0.138 to 0.030.

3.3 Experiment 6: Where new scientific concepts spread next

Artifact: gen_art_experiment_6, iteration 2. Conditional logit on concept-year risk sets from 653 newborn concepts (279 DEV, 248 held-out cohort, 126 held-out field). Retained-field relatedness predicted the next field entered: held-out LR 71.7 ($p = 2 \times 10^{-17}$), $d_0 = 0.302$ [0.240, 0.369], DL pooled 0.284 [0.216, 0.352], $I^2 = 0$, label-permutation $p = 0.001$, rewired-backbone $p = 0.015$. The gateway weighting added nothing (M3 vs M1 permutation $p = 0.17$). Trajectory clustering by DTW gave $k = 2$ stable classes (bootstrap ARI 1.0) but the HMM ($S = 6$) did not reproduce it (ARI 0.094).

Table 5: Experiment 6: field-entry conditional logit and held-out decisions (gen_art_experiment_6, iteration 2).

Metric	Value
DEV d_0	+0.250 [+0.182, +0.321]
DEV LR (M2 vs M0)	38.6 ($p = 5.1 \times 10^{-10}$)
DEV permutation p	0.009
DEV rewired-backbone p	0.030
DEV AUC M0 / M2	0.801 / 0.805
H2 entry CONFIRMED	all 6 clauses pass
H2 ordering CONFIRMED	$p_{\text{gw}} = 0.655$, sign $p = 0.003$
Rescue SUPPORTED	No ($R_1 = -0.217$, CI incl. zero)
Relay SUPPORTED	No ($\beta_{\text{ret} \times \text{gate}} = -1.30$, CI incl. zero)

3.4 Evaluation 1: Stress-testing the gateway-field retention lead (iteration 2)

Zero-API stress test of the iteration-1 lead. The union panel (362 de-duplicated episodes, 54 concepts) gave $\Delta\text{AUC} +0.001$ [−0.012, +0.012]. The shuffled- R placebo’s 95th percentile (0.130) exceeded the original +0.103, so the original lead could not be certified as above chance on 80 episodes.

Table 6: Evaluation 1: gateway retention ΔAUC across panels (gen_art_evaluation_1, iteration 2).

Panel	ΔAUC over M2	95% CI	Episodes
Exp4 (iteration 1)	+0.037	$[-0.018, +0.130]$	80
Exp1 (s2-fos crosswalk)	+0.001	—	367
Exp3	−0.006	—	129
Union (de-duplicated)	+0.001	$[-0.012, +0.012]$	362
New-episodes-only	−0.001	$[-0.021, +0.017]$	282
DL pooled	+0.0015	$(I^2 = 0)$	—

3.5 Dataset 2: External recognition lookup (gen_art_dataset_2, iteration 2)

External-recognition lookup table (outcome O5) for all 65,026 OpenAlex legacy concepts, built with zero credits and \$1.49 of OpenRouter calls. Sources: MeSH 2026 (20,872 concepts), English Wikipedia creation dates (6,540 exact first revisions), Wikidata P571/P575 (1,425), ACM CCS, MSC, PACS/PhySH, curated breakthrough lists (589 concepts), and JEL. Quality: audit precision 0.96 for label matches. 67% of concepts were recognised at or before t_0 . O5 was built but proved unrelated to publication outcomes (pooled ρ with $O_{2r} = 0.014 [-0.045, +0.073]$).

3.6 Research 1: Positioning study (gen_art_research_1, iteration 2)

Positioning for the Applied Network Science paper. 22 citable ANS papers identified. Comparison numbers: Guevara 2016 field-entry AUC 0.896/0.715/0.682 (individuals/organisations/countries). Weng 2013: $\sim 7\times$ precision of random guessing from the first 50 tweets. Adopter-centrality retention was judged NEW for concept adoption by fields but partially anticipated in general.

Table 7: Research 1: novelty verdicts (gen_art_research_1, iteration 2).

Claim	Verdict
Position-based retention (concept adoption by fields)	NEW
Relay mechanism (gateway fields relay onward)	Partially anticipated
Rescue-and-relay mechanism	Partially anticipated
Leave-concept-out propensity control	NEW
Concept $\times$ field episode unit	Partially anticipated

3.7 Dead ends (iteration 2)

- **Gateway retention (H1):** DISCONFIRMED on 27,393 episodes ($\Delta\text{AUC} -0.00001$). The signal vanished once the field’s own retention propensity was controlled.
- **Gateway weighting of entry:** added nothing (permutation $p = 0.17$).
- **O1 gains for G variants:** all 8 were label-coverage artefacts ($G +0.072 \rightarrow +0.002$).
- **Two-class trajectory typology:** DTW vs HMM ARI 0.094; the localised class was dominated by Medicine homes.

- **O5 external recognition:** unrelated to any indicator or model beyond B5 + onset year.

3.8 Hypothesis update after iteration 2

The headline moved from gateway centrality (closed) to the retained-frontier lead from Experiment 6 ($d_0 = 0.28$, one frame). The reviewer’s nearest-neighbour objection (standard RCA-thresholded density) became the decisive test for iteration 3. Evidence state: lead. Confidence: decreased. The review scored at 3 (blocking).

4 Iteration 3: The Decisive Tests

4.1 Why this iteration ran

Iteration 3 aimed to deliver the validated framework: (RQ1) which temporal network indicators anticipate emergence beyond count baselines, validated on held-out fields; (RQ2) the retained-frontier claim tested against the conventional RCA-thresholded density rival. Five artifacts were commissioned: Experiment 7 (decisive frontier test), Experiment 8 (53-indicator held-out screen), Experiment 9 (RQ2 trajectories), Evaluation 2 (record audit), and Research 2 (prior-art check). Four completed; Experiment 9 failed (output-format validation error; method.py was never run).

4.2 Experiment 7: Retained-frontier decisive test (iteration 3)

An independent frame was built: Experiment 5’s 12,499 concepts minus every Experiment 6 concept by ID/QID/normalised label, giving 11,841 concepts. The spec was hash-frozen on DEV (4,486 concepts); held-out was scored once.

Table 8: Experiment 7: retained-frontier coefficient d_0 across held-out groups, independent frame (gen_art_experiment_7, iteration 3). LR = likelihood ratio test of R3 vs R2.

Group	d_0	95% CI	n concepts
Physical Sciences	+0.148	[+0.074, +0.219]	656
Life & Environment	+0.401	[+0.347, +0.458]	1,071
Social Sciences	+0.297	[+0.245, +0.345]	1,274
Math & Decision	+0.065	[−0.110, +0.234]	161
Pooled held-out	+0.322	[+0.291, +0.355]	3,162
DL pooled (4 groups)	+0.243	[+0.118, +0.368]	($I^2 = 0.92$)
Cohort 2010–14	+0.321	[+0.292, +0.347]	—
LR (R3 vs R2), held-out		325.8 ($p < 10^{-70}$)	

d_0 survived both the conventional RCA > 1 density (D_{rca}) and a share-weighted current-presence density: adding d_0 after both rivals gave LR = 29.3 ($p = 6.1 \times 10^{-8}$) on DEV, and D_{rca} was absorbed once d_0 entered. All three specificity tests rejected their nulls after Holm correction.

FRONTIER = PARTIAL. The pre-declared volume-matched contrast was null: -0.028 $[-0.105, +0.046]$ on held-out. The effect was backbone-specific: under minimum conditional probability proximity (Hidalgo), $d_0 = -0.021$ ($p = 0.012$). Dose by retention age: 2 years +0.098; 3 years +0.075; ≥ 4 years +0.304 (not monotone on held-out).

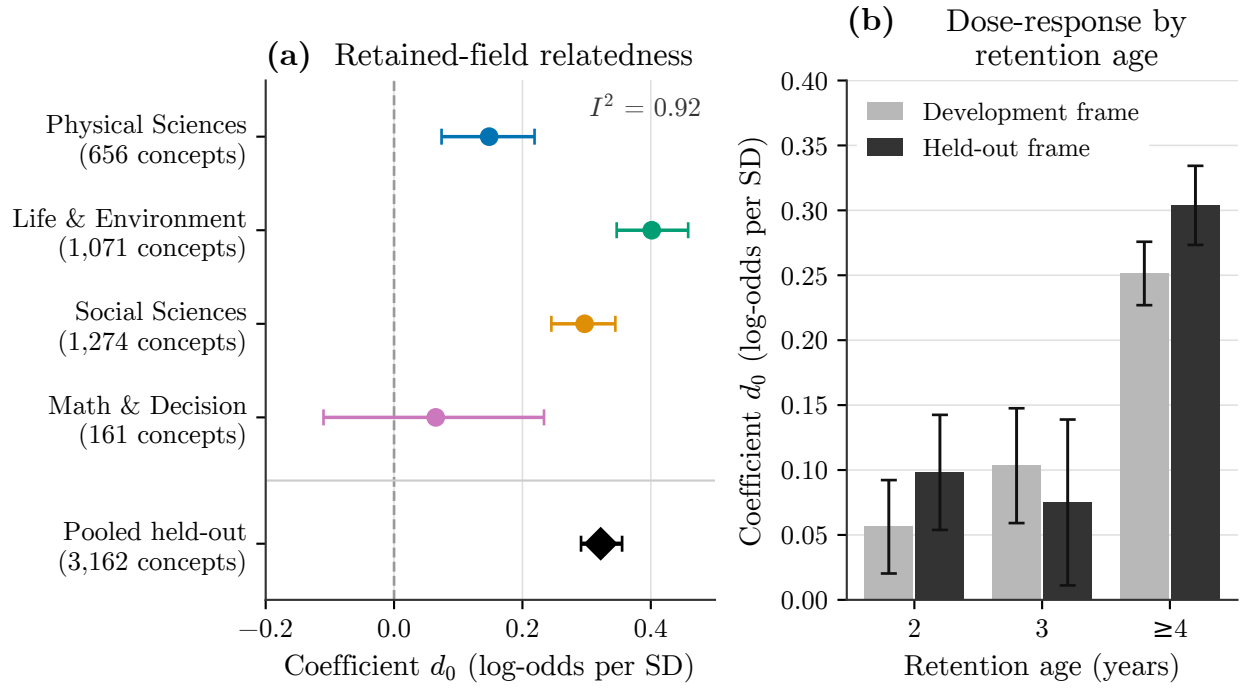


Figure 3: Retained-field relatedness predicts the next field a concept enters. (a) Conditional-logit coefficient d_0 on the held-out groups of the independent frame. Coloured circles are per-group estimates with 95% concept-bootstrap CIs. The black diamond is the pooled held-out estimate, $d_0 = 0.322$ [0.291, 0.355] (3,162 concepts, 6,978 entry events). Three groups are clearly positive. The Math & Decision interval crosses zero. (b) d_0 split by retention age (2, 3, ≥ 4 years).

4.3 Experiment 8: 53-indicator held-out screen (iteration 3)

RQ1 held-out deliverable on the 12,499-concept frame. 53 indicators in 7 families over $t_0..t_0 + 2$ (popularity E, disciplinary F, landing G, retained-frontier FR, 27 co-occurrence ego-network A, co-author S) plus B5 baseline. DEV-only ranking frozen, hash sealed, single unseal, DL pooling, Holm correction.

Table 9: Held-out indicator screen: the 10 indicators with highest DEV partial Spearman priority for rarefied breadth (O_{2r} , $m = 50$), tested on held-out groups (gen_art_experiment_8, iteration 3). DerSimonian–Laird pooled estimates with 95% CI.

Indicator	Family	Pooled PSP	95% CI	I^2	Sign	Confirmed
M0_density_end	FR	+0.375	[+0.279, +0.462]	0.74	6/6	Yes
D_vol_end	FR	+0.307	[+0.256, +0.356]	0.10	6/6	Yes
CONTACT	FR	+0.211	[+0.161, +0.261]	0.00	6/6	Yes
_REACH						
n_{comm} (W3)	A	+0.167	[+0.063, +0.267]	0.78	6/6	Yes
NOV _{res}	A	+0.151	[+0.044, +0.255]	0.75	6/6	Yes
RETENTION	FR	−0.114	[−0.160, −0.067]	0.00	6/6	Yes
_RATIO						
ego_density (W3)	A	−0.102	[−0.151, −0.053]	0.00	6/6	Yes
Rao–Stirling	A	−0.072	[−0.153, +0.010]	0.44	5/6	No
G_{btw}	G	+0.056	[−0.006, +0.118]	0.33	6/6	No
log offhome vol.	E	−0.089	[−0.171, −0.007]	0.63	5/6	No

Seven of ten survived held-out validation (Holm $p < 0.05$, CI excluding zero): five positive and two negative. No centrality or volume-only indicator survived. An ElasticNet combining all indicators added Spearman +0.059 [+0.046, +0.073] over B5 on held-out groups; an Explainable Boosting Machine added +0.052 [+0.037, +0.067].

Pre-onset footprint caveat. M0_density_end and D_vol_end used cumulative field history from 1995 to $t_0 + 2$. About half of their signal came from the concept’s pre-onset field footprint: postonset-only field density dropped from +0.374 to +0.187 [0.145, 0.246], and volume-weighted density from +0.317 to +0.176 [0.114, 0.227].

4.4 Evaluation 2: Record audit (gen_art_evaluation_2, iteration 3)

Audited 246 claims in the ledger: 224 MATCH, 15 MISLABELLED, 6 MISMATCH, 1 FILE_FLAG_OVERRIDDEN; 58 blocking items. Produced ready-to-paste corrections covering H1 criteria, ordering (reclassified as MIXED), H3 CI including zero, Dataset 2 coverage counts, and many iteration-1/2 items.

Frame agreement between Experiment 5 and Experiment 6: 628 shared concepts, onset ± 1 agreement ≥ 0.80 , home $\kappa \geq 0.60$, O_{2r} Spearman ≥ 0.70 ; retention κ failed at 0.48 (matched definition: 0.98). O5 readings: O5_main UNRELATED, O5_wiki UNRELATED, O5_tax RELATED_NOT_DUPLICATE. O5 hand-check precision: strict 0.86, lenient 0.96 ($n = 100$, FIT_FOR_USE = true).

4.5 Research 2: Prior art and venue check (gen_art_research_2, iteration 3)

The retained-frontier claim (Claim A) was judged PARTIALLY ANTICIPATED: persistence had been used only as a filter on the entry outcome (Pinheiro 2022, Albora 2023), and no retained-only

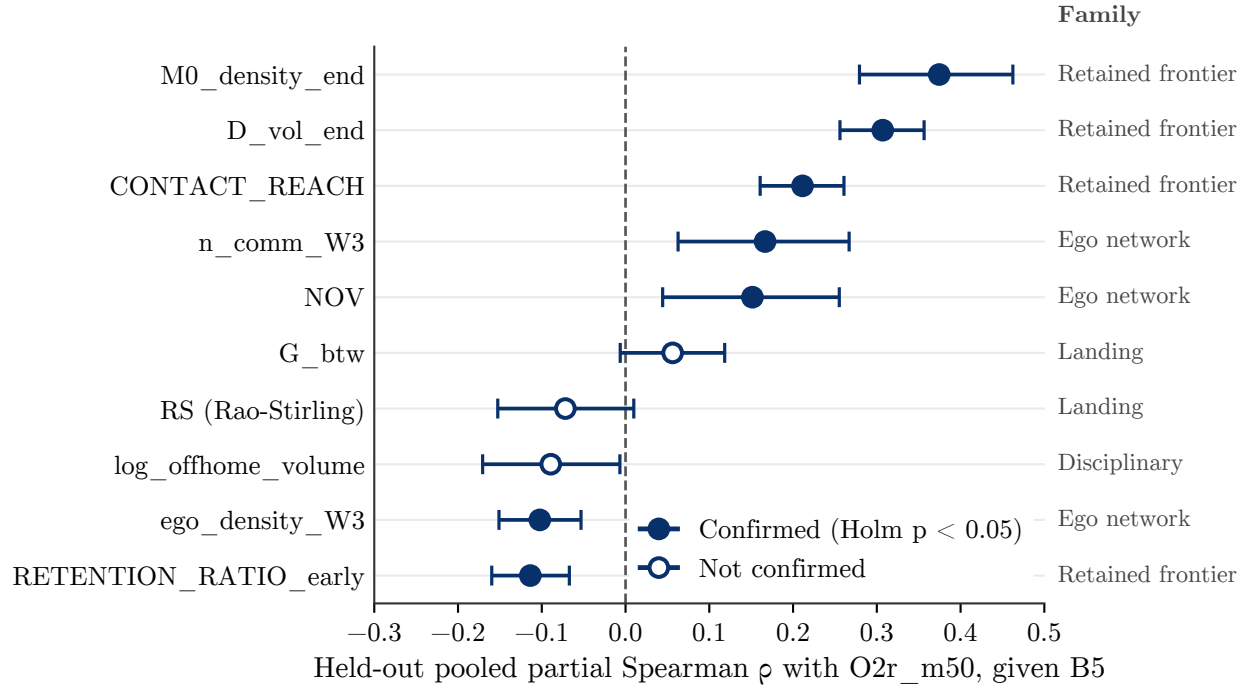


Figure 4: Held-out screen of the 10 indicators frozen on the DEV home groups for rarefied cross-field breadth. Each row shows the DerSimonian–Laird pooled partial Spearman ρ with 95% CIs. Filled circles mark confirmed indicators; open circles mark those not confirmed.

Table 10: Evaluation 2: claims ledger status counts (gen_art_evaluation_2, iteration 3).

Status	Count
MATCH	224
MISLABELLED	15
MISMATCH	6
FILE_FLAG_OVERRIDDEN	1
Total claims audited	246
Blocking items	58

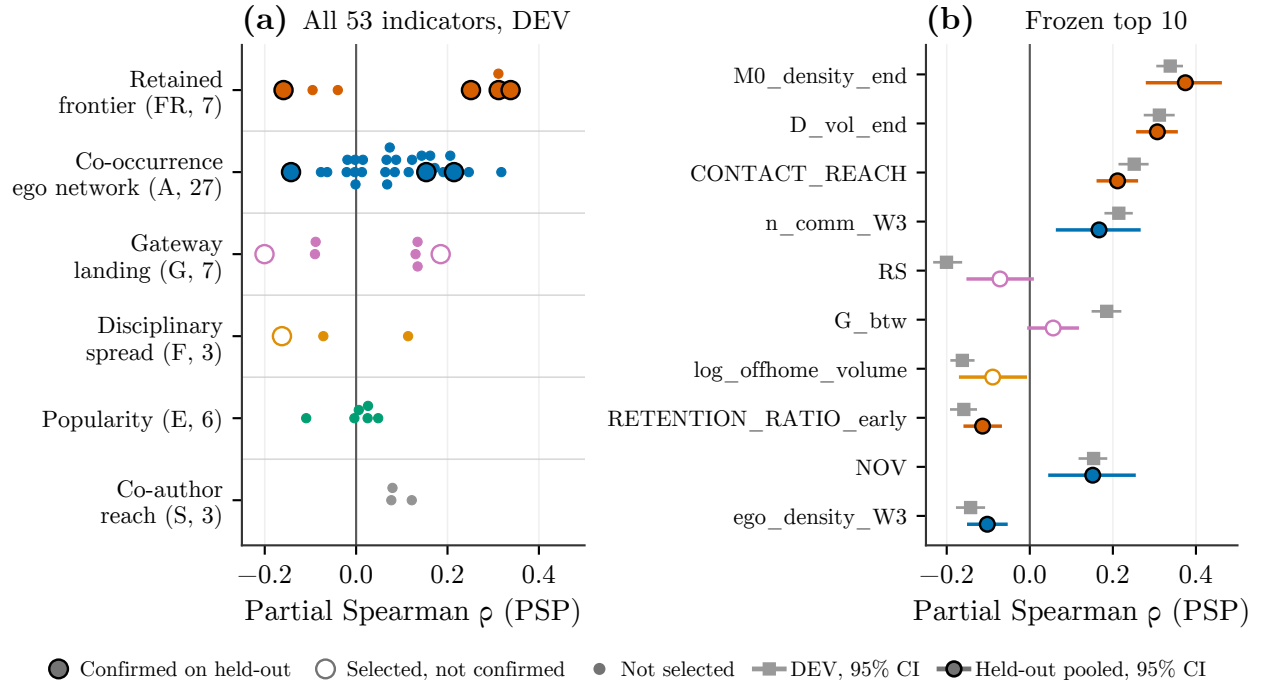


Figure 5: Complete indicator screen for rarefied cross-field breadth. (a) Partial Spearman ρ of all 53 screened indicators on the DEV fields. Colour marks the family. Large filled circles are the 7 confirmed indicators; large open circles are the 3 selected but not confirmed; small dots are the 43 not selected. (b) The frozen top 10 in DEV rank order, with DEV and held-out estimates.

or duration-weighted density predictor was found. Claim B (relatedness to dropped fields lowers entry) was judged mechanism partly anticipated (Fernandes & Tang 2014) but NEW as a test.

Table 11: Research 2: novelty verdicts and missing rivals (gen_art_research_2, iteration 3).

Item	Verdict
Claim A (retained-frontier entry)	Partially anticipated
Claim B (dropped-field penalty)	NEW as a test
Missing rival R1: persistence-filtered D_{rca}	Not tested yet
Missing rival R3: pre-entry intensity/trend	Not tested yet
Missing rival R4: neighbour-momentum density	Not tested yet
RQ1 held-out indicator screen	No comparable prior
RQ2 breadth decomposition	NEW

4.6 Dead ends (iteration 3)

- **gen_art_experiment_9** (RQ2 trajectories on 12k concepts): failed. Output-format validation error; method.py was never run.
- **O5 external recognition:** pooled ρ with $O_{2r} = 0.014$ $[-0.045, +0.073]$. No indicator or model beat B5 + onset year.
- **Abandonment penalty:** d_{lost} in the additive model (A1) = -0.007 $[-0.036, +0.022]$; null.
- **Candidate S:** tested in Experiment 8 (S_{comp} , S_{comp_n} , $S_{\text{isolated_share}}$); not confirmed for any outcome.

4.7 Hypothesis update after iteration 3

The headline moved from the retained frontier (PARTIAL) to openness vs consolidation from Experiment 8: new_edge_rate +0.118, n_{comm} +0.167, participation +0.150, NOV_{res} +0.139, ego_density -0.102 , edge_persistence -0.080 . A fresh 2015–17 onset cohort was declared for confirmation. Evidence state: lead. Confidence: decreased. Review score: 3 (blocking).

5 Iteration 4: Fresh-Cohort Test and RQ2 Decomposition

5.1 Why this iteration ran

Iteration 4 attacked the Experiment 8 openness lead on four sides: replication on a never-screened fresh cohort (Experiment 10), within-concept mechanism via closure test (Experiment 11), RQ2 trajectories and decomposition on 12k concepts (Experiment 12), boundary study plus record corrections (Evaluation 3), and prior-art check (Research 3). Five artifacts were commissioned; four completed. Experiment 11 sealed a spec and ran DEV models but was interrupted before held-out scoring.

5.2 Experiment 10: Fresh-cohort OPEN test (iteration 4)

Single-unseal confirmation on a fresh 2015–2017 onset cohort of 1,443 concepts (573 with $\text{OPEN}_{\text{home}}$) that no earlier screen had touched. OPEN is the mean of six signed, z -scored ego-network components, with constants frozen on the 12,499 Experiment 5 concepts.

Table 12: OPEN index control ladder on the 2015–2017 confirmatory cohort (gen_art_experiment_10, iteration 4). PSP = partial Spearman priority with rarefied breadth (O_{2r} , $m = 50$), 95% concept-bootstrap CI.

Build	R0	R1	R2	R3	R4	R5	n
$\text{OPEN}_{\text{home}}$	+.12	+.10	+.09	+.08	+.07	+.06	573
OPEN_{all}	+.21	+.18	+.17	+.17	+.15	+.14	630
$\text{OPEN}_{\text{size}}$	+.18	+.15	+.15	+.14	+.12	+.11	591

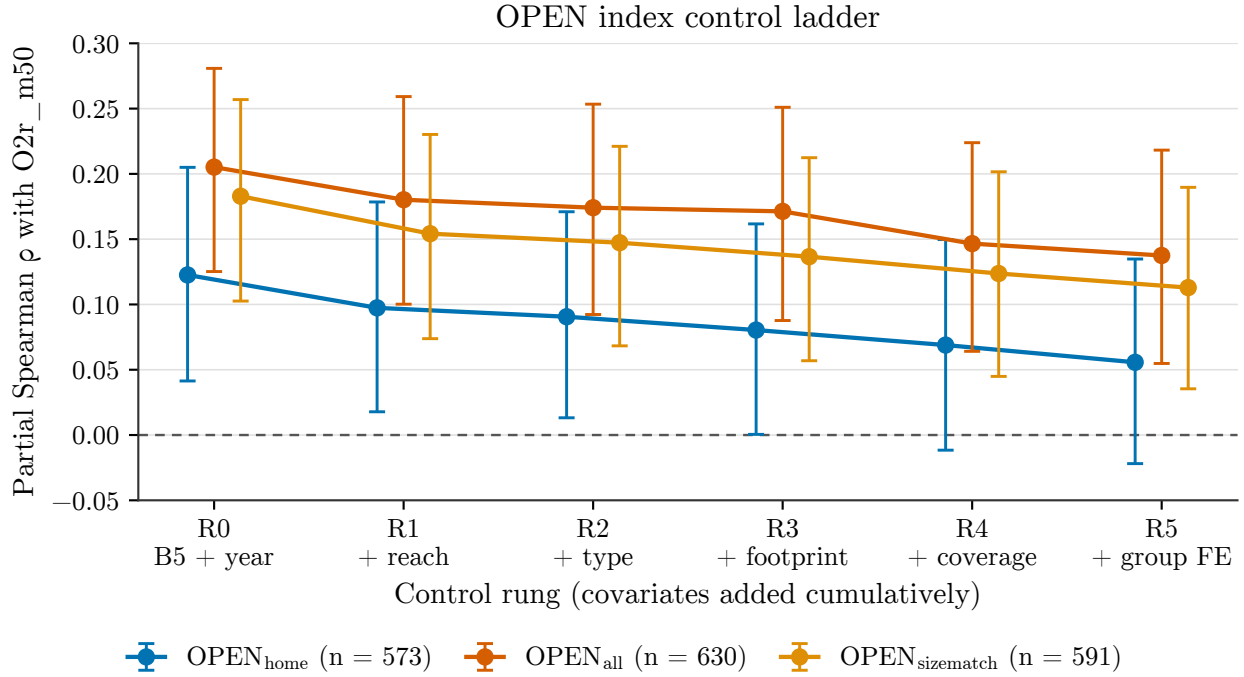


Figure 6: Control ladder for the early ego-network openness index (OPEN) on the fresh 2015–2017 onset cohort. The x-axis gives cumulative control rungs. Circles are point estimates; error bars are 95% concept-bootstrap CIs ($B = 2,000$); the dashed line marks $\rho = 0$.

Frozen verdict: CONFIRMED but marginal. $\text{OPEN}_{\text{home}}$ partial Spearman with O_{2r} was +0.091 [+0.013, +0.171] at R2 and +0.080 [+0.001, +0.162] at R3. CIs included zero at R4/R5, and the DL pool over groups was +0.083 [−0.007, +0.173]. It added no practical prediction: B5 Spearman 0.768 vs 0.770 (+0.002 [−0.003, +0.008]).

Mechanical coupling was large: OPEN_{all} +0.174; ALL minus HOME +0.093 [+0.016, +0.169].

5.3 Experiment 11: Within-concept closure test (iteration 4, incomplete)

Hash-sealed preregistration. DEV body models on 35,328 concept-years from 4,661 concepts. Verdict: NOT SUPPORTED. Density PPML $b = -0.070$ $[-0.180, +0.040]$; OPEN_{home} $b = +0.015$ $[-0.038, +0.069]$. The Sun–Abraham event study was interrupted (OpenBLAS error). Held-out and cohort were not scored.

5.4 Experiment 12: Breadth decomposition (iteration 4)

Log-additive decomposition of the breadth gap on all 12,499 concepts.

Table 13: Breadth decomposition ($\log B_n = \log E_2 + \log M + \log \rho$): share of the top-versus-bottom tercile gap in rarefied breadth attributable to each component (gen_art_experiment_12, iteration 4). DEV concepts, $n = 3,188$.

Component	Share	95% CI
Exploration ($s_{E_2} + s_M$)	0.732	[0.703, 0.764]
Early contact diversity (s_{E_2})	0.779	[0.738, 0.818]
Frontier advance (s_M)	-0.046	$[-0.074, -0.017]$
Retention (s_ρ)	0.268	[0.236, 0.297]
Difference (explore – retain)	0.464	[0.407, 0.528]

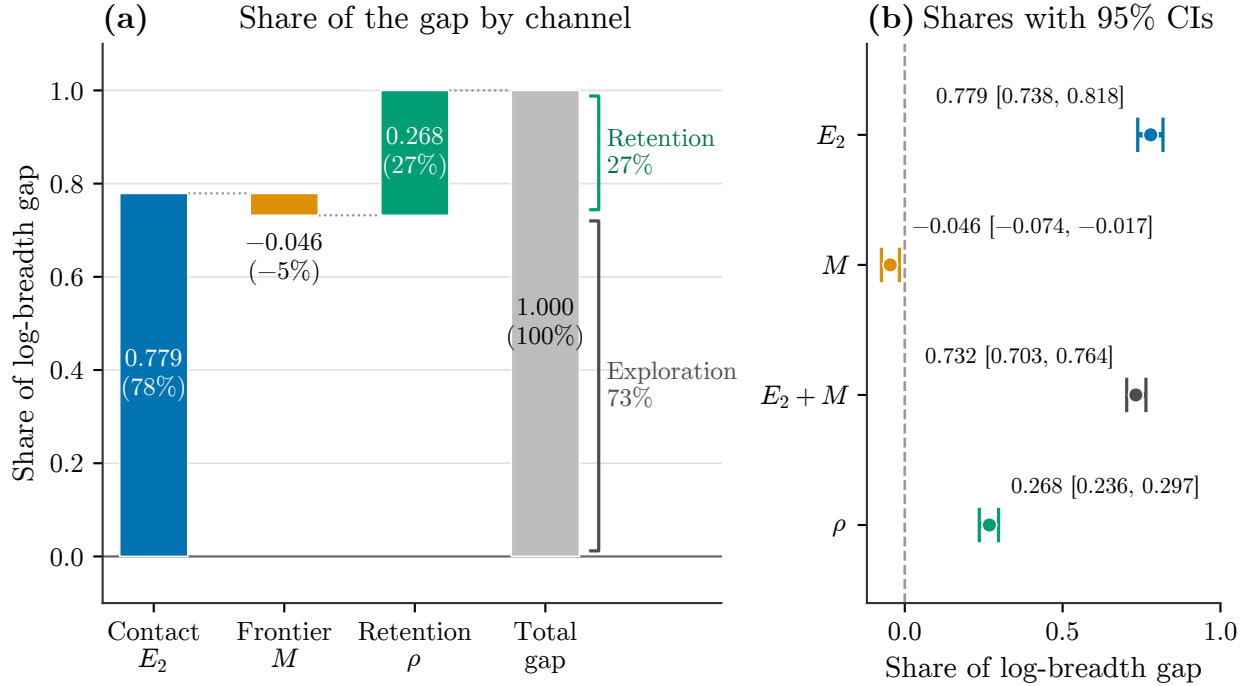


Figure 7: Log-additive decomposition of the gap in retained off-home breadth between the top and bottom terciles of size-adjusted breadth. Early contact diversity E_2 (blue) contributes 78%, frontier advance M (amber) is slightly negative at -5%, and retention ρ (green) contributes 27%.

No trajectory typology passed the naming rule (DTW $k = 4$ vs HMM $S = 5$ ARI 0.222; held-out re-cluster ARI 0.44/0.38). Verdict: CONTINUUM along a PC1 breadth-of-spread axis (38.8%) and

a PC2 keep-vs-lose axis (10.7%).

5.5 Evaluation 3: Boundary study and corrections pack

Artifact: gen_art_evaluation_3, iteration 4.

Post-onset rescore: M0_density_end +0.374 $\rightarrow$ +0.187 post-onset (attenuation 0.50); D_vol_end +0.317 $\rightarrow$ +0.176 (0.45). OPEN pooled PSP +0.181 [0.082, 0.277] (DL over 4 held-out groups plus two cohort parts, 6/6 positive). Specification curve: 99.7% of 1,920 specifications with CIs above zero (permutation $p = 0.005$). Claims ledger v3: 1,290 rows, 0 MISMATCH, 0 NOT_FOUND.

Table 14: Evaluation 3: boundary and specification-curve results (gen_art_evaluation_3, iteration 4).

Metric	Value
M0_density_end full PSP	+0.374
M0_density_end post-onset PSP	+0.187 [0.145, 0.246]
M0_density_end attenuation	0.50
D_vol_end full PSP	+0.317
D_vol_end post-onset PSP	+0.176 [0.114, 0.227]
D_vol_end attenuation	0.45
OPEN DL4 PSP	+0.181 [0.082, 0.277]
OPEN DL6 PSP	+0.163 [0.108, 0.218]
OPEN DL4/6 sign positive	6/6
Spec. curve: specs with CI > 0	99.7% of 1,920
Spec. curve permutation p	0.005
Ledger v3 rows / MISMATCH	1,290 / 0

5.6 Research 3: Prior-art check (gen_art_research_3, iteration 4)

C1 (openness $\rightarrow$ breadth): PARTIALLY ANTICIPATED. The same direction shown for concept pairs (Maillart 2026), papers (Wang 2017), memes (Weng 2013) and people (Ugander 2012). No prior study combined the concept unit, a size-adjusted breadth outcome and held-out fields. C2 (consolidation $\rightarrow$ less breadth): PARTIALLY ANTICIPATED in mechanism. Cheng et al. 2023: ideational consistency predicts volume (+53%/SD) but the authors did not study cross-domain translation.

Table 15: Research 3: novelty verdicts for the openness claims (gen_art_research_3, iteration 4).

Claim	Verdict
C1: openness $\rightarrow$ breadth	Partially anticipated
C2: consolidation $\rightarrow$ less breadth	Partially anticipated
C3: low retention ratio $\rightarrow$ breadth	NEW
C4: within-concept closure precedes entry slowdown	NEW

5.7 Dead ends (iteration 4)

- **Experiment 11 (within-concept closure):** NOT SUPPORTED on DEV. Density -0.070 , OPEN $+0.015$, both CIs include zero.
- **Typology:** DTW-HMM ARI 0.222; no discrete classes pass the naming rule. Continuum.
- **RETENTION_RATIO_early:** null on the fresh cohort at R2/R3.
- **Frontier advance:** slightly negative ($s_M = -0.046$).

5.8 Hypothesis update after iteration 4

Claim sharpened from six-component “openness” to the decoupled home-only signal: novel partners (NOV_{res} $+0.134$) and churn (edge_persistence -0.112). OPEN_{home} was marginal (DL CI includes zero, no forecasting gain). A second vocabulary-free population (Frame N) was declared for final confirmation. Evidence state: lead. Confidence: decreased. Review score: 2 (blocking, due to fabricated case-study rows and missing Experiment 11).

6 Iteration 5: Final Tests and Evidence Synthesis

6.1 Why this iteration ran

The final iteration tested the home-field churn/novelty signal on a second, vocabulary-free population (Frame N), ran the Cheng consistency reversal, completed the Exp11 closure test, checked whether “churn” was sampling noise, and repaired the record with an evidence synthesis. Five artifacts completed: Experiment 13 (Frame N), Experiment 14 (Cheng reversal), Experiment 15 (mechanism and Exp11 completion), Evaluation 4 (record repair and evidence pool), and Experiment 16 (confound checks).

6.2 Experiment 13: Frame N vocabulary-free confirmation

Artifact: gen_art_experiment_13, iteration 5. 636 vocabulary-free phrase-born concepts mined from random title samples (2003–2015 onsets), excluding all legacy labels. These newborns had 14% lower breadth and 89% higher transience than legacy concepts, confirming that curated vocabularies are survivor-selected.

Frozen verdict: PARTIAL. OPEN_{home} PSP $+0.117$ [$+0.020$, $+0.218$] at R3, $+0.086$ [-0.009 , $+0.190$] at R5. NOVCHURN_{home} at R3 was $+0.108$ [$+0.007$, $+0.211$]. NOV_{res} carried the signal ($+0.208$ [$+0.113$, $+0.303$]); edge persistence was null (-0.013). 3 of 4 estimable groups positive (SOC -0.025); DL $+0.112$ [-0.015 , $+0.239$]; Holm $p = 0.052$.

6.3 Experiment 14: Cheng consistency reversal (iteration 5)

Rebuilt Cheng et al. (2023) ideational consistency on OpenAlex topic co-usage for 12,311 concepts (105,839 concept-years).

Adding current volume removed $\sim 98\%$ of the consistency–volume effect. As an early trait, consistency predicted *narrower* cross-field reach: pooled PSP -0.069 [-0.093 , -0.047] (DL over 5 groups: -0.079 , $I^2 = 0$, 5/5 negative). Within concepts, a more consistent year was followed by slightly more entries ($b = +0.025$ [0.004 , 0.048]), so the reach penalty was a between-concept trait.

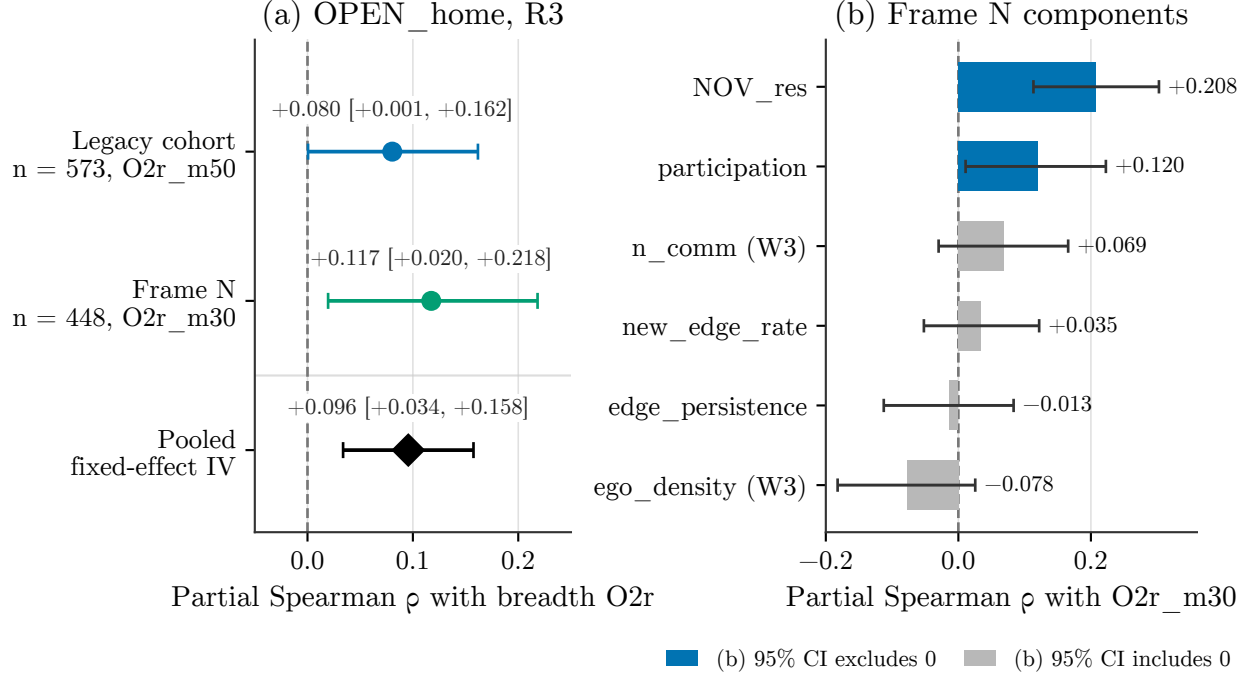


Figure 8: Confirmation of the OPEN signal on vocabulary-free Frame N concepts. (a) Forest plot of the $\text{OPEN}_{\text{home}}$ partial Spearman at the footprint rung (R3): the 2015–17 legacy cohort gives +0.080 [+0.001, +0.162], Frame N gives +0.117 [+0.020, +0.218], and their inverse-variance pool gives +0.096 [+0.034, +0.158]. (b) PSP for the six OPEN components on Frame N. NOV_{res} carries the signal (+0.208); edge persistence is null (−0.013).

Table 16: Cheng consistency reversal (`gen_art_experiment_14`, iteration 5). Test A: effect on next-year volume (%/SD). Test B: partial Spearman with rarefied breadth (O_{2r} , $m = 50$) given B5.

	Volume (%/SD)	Breadth PSP
NB twin (Cheng design)	+53.5%	—
PPML	+83.1%	—
PPML + $\log V(t)$	+1.3% [+0.5, +2.1]	—
A2/A1 ratio	0.021 [0.009, 0.035]	—
Pooled (5 groups, DL)	—	−0.079 ($I^2 = 0$)
All 5 groups negative	—	5/5
2015–17 cohort	—	−0.111 [−0.197, −0.030]
Spearman with edge persistence	—	+0.77

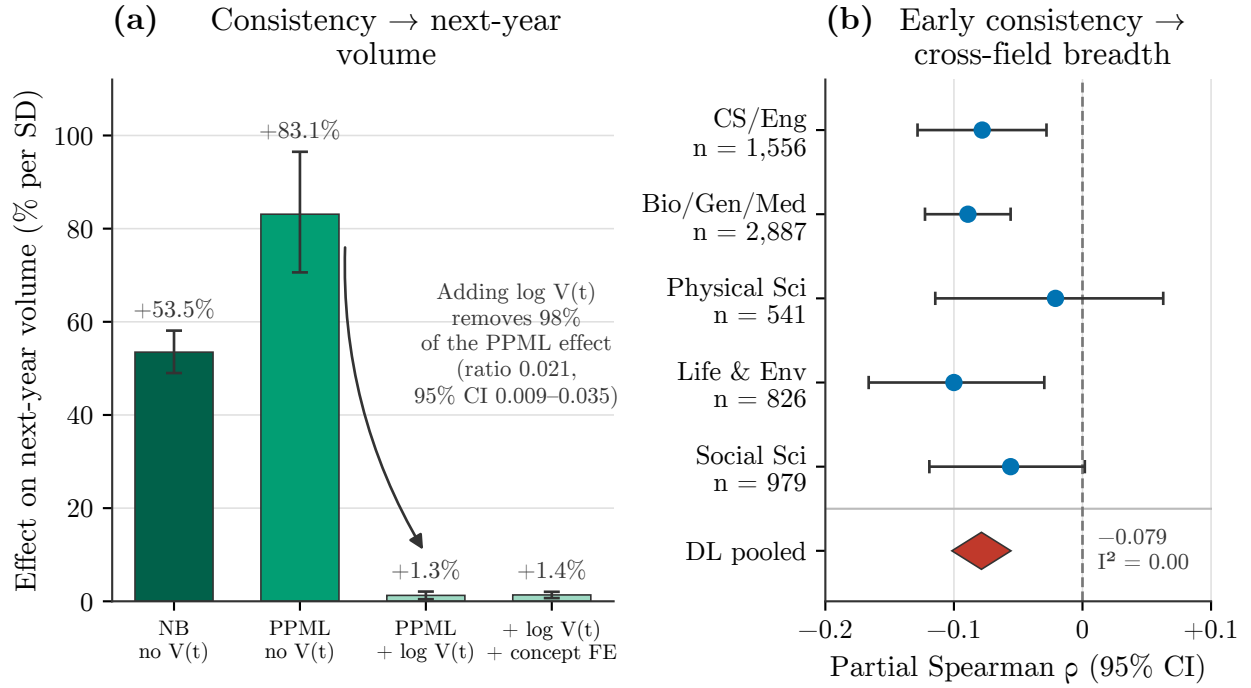


Figure 9: The consistency–breadth reversal. (a) Effect of ideational consistency on next-year volume: +53.5% per SD without volume control, dropping to +1.3% once current volume is added (98% removed). (b) Partial Spearman ρ between early consistency and rarefied breadth given B5: DL pooled -0.079 $[-0.102, -0.056]$, $I^2 = 0$. All five point estimates are negative.

6.4 Experiment 15: Mechanism and Exp11 completion

Artifact: gen_art_experiment_15, iteration 5.

Partner decomposition. The home-only signal was carried by partners from new Leiden communities ($C_2 = +0.102$ [$+0.069, +0.133$], Holm $p = 0.003$) that arrived through mixed-field papers ($C_4 = +0.103$ [$+0.071, +0.134$], Holm $p = 0.003$). Domain novelty was the main contributor (Shapley $\phi = +0.132$; method = $+0.012$). Bridging papers (5% of early home papers) halved NOVCHURN’s PSP ($0.118 \rightarrow 0.056$).

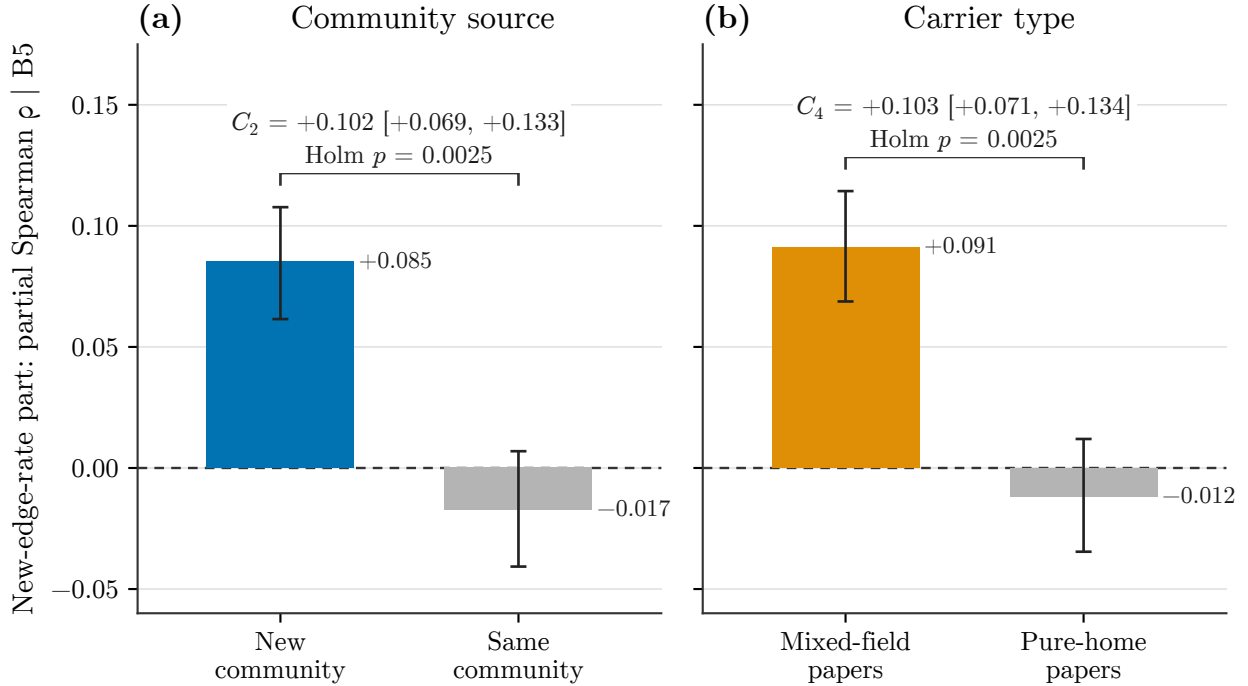


Figure 10: Which co-occurrence partners carry the openness–breadth signal? (a) Partners from new communities carry the signal (contrast $C_2 = +0.102$, Holm $p = 0.0025$); same-community partners do not. (b) Partners arriving through mixed-field papers carry it ($C_4 = +0.103$, Holm $p = 0.0025$); partners from pure-home-field papers do not.

Exp11 completion. The sealed code was completed with a path-only patch and single BLAS threads. DEV verdict unchanged: NOT SUPPORTED. Old held-out PPML: density $+0.068$ [$-0.072, +0.209$]; $\text{OPEN}_{\text{home}} -0.079$ [$-0.146, -0.013$], the opposite of the predicted sign. Cohort 2010–14: both null.

Table 17: Experiment 15: cross-validated Spearman gain from NOVCHURN and OPEN over B5, by body (gen_art_experiment_15, iteration 5).

Body	n	B5 ρ	B5+NOVCHURN ρ	Gain [95% CI]
DEV	3,188	0.761	0.764	+0.003 [+0.001, +0.005]
Old held-out	1,833	0.706	0.710	+0.004 [+0.001, +0.007]
Cohort 2010–14	2,182	0.764	0.766	+0.001 [−0.001, +0.004]
Cohort 2015–17	634	0.785	0.788	+0.003 [−0.002, +0.008]

Trait stability. Yearly $\text{OPEN}_{\text{home}}$ ICC was 0.37–0.39 (REML agreed). NOVCHURN was

0.26–0.29. The window retest was 0.51–0.57; disattenuated retest 0.86–0.91.

6.5 Experiment 16: Confound checks (iteration 5)

Table 18: Confound-correction variants of NOVCHURN (gen_art_experiment_16, iteration 5). Pooled PSP with O_{2r} ($m = 50$) given B5+R2 covariates.

Variant	Pooled PSP	Retention	n
NOVCHURN _{raw}	+0.116	—	6,203
V2 excess (year-permutation)	+0.008	6.5%	6,203
V1 fixed- n rarefaction ($n = 10$)	+0.078	68%	2,872
V3 configuration z -score	+0.106	100%	3,993
Chao-2005 Jaccard	+0.106	91%	—
OPEN _{home, clean}	+0.115	125%	6,322

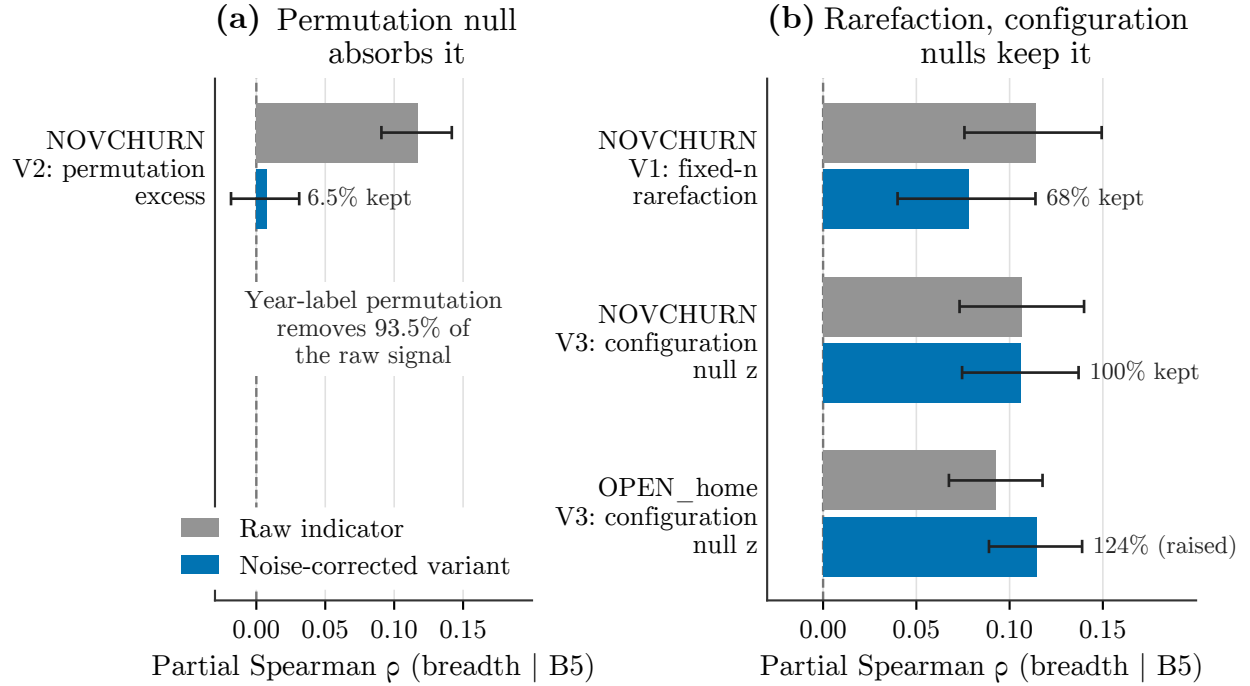


Figure 11: The churn/openness signal behaves like topical non-redundancy of the home neighbourhood, not year-to-year partner turnover. (a) The within-concept year-label permutation excess cuts NOVCHURN from +0.117 to +0.008, keeping 6.5%. (b) Fixed- n rarefaction keeps 68%; configuration-null z -scores keep 100% and raise OPEN_{home} to 125%.

The within-concept permutation null absorbed the year-to-year churn: excess NOVCHURN pooled PSP = +0.008 [−0.02, +0.03], compared to raw +0.116 [+0.09, +0.14]. The R^2 of raw persistence on this permutation-null mean was 0.66. What predicted breadth was a static property—the non-redundancy of the concept’s home-topic partner mix—not temporal partner turnover. However, V2 excess variants had split-half reliability of ~ 0.01 – 0.05 , and the planted-churn check failed, so V2 could not adjudicate temporal churn at ~ 10 papers/year.

Split-half reliability: $\text{NOVCHURN}_{\text{raw}}$ 0.48, $\text{OPEN}_{\text{home}}$ 0.49, $\text{OPEN}_{\text{home, clean}}$ 0.58. Disattenuated pooled $\text{NOVCHURN}_{\text{raw}}$ PSP: 0.178 [0.141, 0.214].

6.6 Evaluation 4: Record repair and evidence synthesis (iteration 5)

Ten MUST-FIX items cleared. Claims ledger v4: 1,769 rows, 0 MISMATCH, 0 NOT_FOUND, 0 orphans. Evidence synthesis (descriptive; R2, O_{2r} $m = 50$; DL on Fisher $z + \text{HKSJ}$):

Table 19: Evidence synthesis: $\text{OPEN}_{\text{home}}$ PSP with rarefied breadth across non-selection bodies (gen_art_evaluation_4, iteration 5). DL = DerSimonian–Laird pooled.

Body	PSP	95% CI	n
Held-out PHYS	—	—	742
Held-out LIFEENV	—	—	1,113
Held-out SOC	—	—	1,352
Held-out MATHDEC	—	—	165
Cohort 2010–14	—	—	4,356
Cohort 2015–17	—	—	573
DL pool ($k = 6$)	+0.069	[+0.038, +0.100]	—
HKSJ	+0.069	[+0.042, +0.096]	—
I^2		0	
Bodies positive		6/6	
DEV selection body	+0.109	—	(shrinkage 1.58)
$\text{NOVCHURN}_{\text{home}}$ (DL ($k = 5$))	+0.105	[+0.069, +0.140]	—

6.7 Dead ends (iteration 5)

- **Temporal turnover claim:** the within-concept permutation null absorbed the year-to-year churn. What predicts breadth is a static topical-dispersion property, not temporal partner turnover. However, V2 excess reliability was too low (~ 0.01 – 0.05) to settle the question definitively: “untested rather than refuted.”
- **Frame N Cheng reversal:** not confirmed on Frame N (consistency -0.064 with breadth, CI includes zero).
- **Edge persistence on Frame N:** null (-0.013).
- **$\text{OPEN}_{\text{home}}$ forecasting gain:** effectively zero ($+0.002$ [-0.003 , $+0.008$]).
- **Exp11 held-out:** $\text{OPEN}_{\text{home}}$ coefficient was *negative* on held-out (-0.079), opposite to predicted.

7 What the Run Learned and What Remains Open

7.1 Summary of confirmed findings

RQ1: Indicator screen. Seven of 53 early network indicators survived held-out validation for predicting rarefied cross-field breadth, spanning the retained-frontier and co-occurrence ego-network

families. An ElasticNet combining all indicators added Spearman +0.059 over B5 on held-out groups.

OPEN index. The Eval4 descriptive pool of $\text{OPEN}_{\text{home}}$ over 6 non-selection bodies gave DL partial Spearman +0.069 [+0.038, +0.100], $I^2 = 0$, 6/6 positive. The signal behaves like a static topical-dispersion property; temporal turnover is untested rather than refuted (V2 excess split-half reliability below 0.05).

Consistency–breadth reversal. Cheng et al.’s ideational consistency predicts next-year volume (+53.5%/SD) but net of volume predicts *narrower* cross-field reach (PSP -0.069 [-0.093 , -0.047], 5/5 groups negative, $I^2 = 0$).

RQ2: Retained-field relatedness. On the independent frame (11,841 concepts, 6,978 entry events), concepts spread next to fields related to those currently retaining them: $d_0 = 0.322$ [0.291, 0.355]. However, the volume-matched contrast was null and the coefficient reversed under minimum-conditional-probability proximity.

Breadth decomposition. Early contact diversity accounted for 73% of the breadth gap, with frontier advance near zero. No discrete trajectory typology passed the naming rule.

Mechanism. The home-only signal was carried by partners from new Leiden communities ($C_2 = +0.102$) arriving through mixed-field papers ($C_4 = +0.103$). Domain novelty, not methodological novelty, was the main contributor.

7.2 Effect sizes and practical significance

The practical forecasting gain from early network structure was small. $\text{OPEN}_{\text{home}}$ added Spearman +0.002 over B5 on the fresh cohort. The cross-validated B5 Spearman alone was 0.768; adding all indicators raised it by about 0.03. Split-half reliability of $\text{OPEN}_{\text{home}}$ was 0.49.

7.3 What remains open

- **Temporal turnover vs static non-redundancy:** V2 excess reliability was too low to decide whether temporal churn adds information beyond static topical dispersion.
- **Domain heterogeneity:** Life & Environment showed the weakest OPEN signal; the retained-field relatedness effect had $I^2 = 0.92$.
- **Mechanism:** early ego-network openness could reflect intrinsic generality, community diversity, or problem breadth. Association, not causation, was shown.
- **Vocabulary bias:** Frame N was small (636 concepts), with pre-unseal power of only 0.47.
- **Within-concept closure:** Experiment 11 was null on DEV and opposite-signed on held-out; the event study was incomplete.

7.4 Artifacts produced

The run commissioned 20 artifacts across 5 iterations. 16 completed and 4 failed or were interrupted (gen_art_dataset_1 and gen_art_experiment_2 in iteration 1, gen_art_experiment_9 in iteration 3, and gen_art_experiment_11 partially in iteration 4). All reviews scored 2–3 (blocking), primarily because conclusions in the internal report contradicted the artifacts’ own evidence. The blocking items were corrected over successive iterations, with Evaluation 4 achieving a claims ledger of 1,769 rows with 0 mismatches.